ALLOY DAS

Curriculum Vitae

SCSLab, Department of Mechanical Engineering
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My Webpage
Github

Work Experience

2025 – Present **PhD Research Assistant**, *SCSLab, Department of Mechanical Engineering*, Iowa State University, Ames, IA, USA.
Multi-modal representation learning for 3D plant phenotyping, lighting-robust instance segmentation, and video vs. EEG deep learning for automated rodent seizure detection and severity grading. Advisor: Prof. Soumik Sarkar.

Oct. 2022 – Nov, 2024 **Project Linked Person (Research Assistant)**, *Computer Vision and Pattern Recognition Unit*, Indian Statistical Institute, Kolkata, India.
Gained comprehensive exposure to automatic handwritten answer script evaluation, Scene Text Spotting, and Scene Text Editing.

Sept. 2021 – Mar, 2022 **Research Intern**, *Computer Science & Engineering Department*, National Institute of Technology, Durgapur, India.
Worked on Medical Image Segmentation, Glaucoma Detection, and Covid Detection.

Education

2025–Present **PhD, Mechanical Engineering**, *Iowa State University*, Ames, Iowa, USA.
Advisor: Prof. Soumik Sarkar (SCSLab)

2018 – 2022 **Bachelor of Engineering, Information Technology**, *University Institute of Technology*, Burdwan University, India. CGPA: 8.79/10.
Courses: Artificial intelligence, Image Processing, Machine Learning, Deep learning
Final Year Project: Retinal Disease Glaucoma Segmentation and Detection Using Machine Learning Framework

2015 – 2017 **Higher Secondary Examination**, *T. T. Dr. Bidhan Chandra Roy Vidyalaya (H.S.), West Bengal Council of Higher Secondary Education*, Tribeni, Hooghly, WB, India, Percentage: 74%.

2015 **Secondary Examination**, *T. T. Dr. Bidhan Chandra Roy Vidyalaya (H.S.), West Bengal Board of Secondary Education*, Tribeni, Hooghly, WB, India, Percentage: 86%.

Publications

In Conference Proceedings

2026 Arindam Chowdhury, Rohit Agarwal, **Alloy Das**, Ayush Jalan, Sanket Biswas, and Debashis Nandi. Deep Entropy-Sampled O-Net for Joint Retinal Cup-Disc Segmentation in Glaucoma Detection. In *Smart Innovation, Systems and Technologies*, pages 533–548. Springer, 2026.

2025 **Alloy Das**, Sanket Biswas, Prasun Roy, Subhankar Ghosh, Umapada Pal, Michael Blumenstein, Josep Lladós, and Saumik Bhattacharya. FASTER: A Font-Agnostic Scene Text Editing and Rendering Framework. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 1944–1954, 2025.

2025 Aniket Pal, Sanket Biswas, **Alloy Das**, Ayush Lodh, Priyanka Banerjee, Soumitri Chattopadhyay, Ajoy Mondal, Dimosthenis Karatzas, Josep Lladós, and C. V. Jawahar. ICDAR 2025 Handwritten Notes Understanding Challenge. In *International Conference on Document Analysis and Recognition (ICDAR), Lecture Notes in Computer Science*, pages 553–567. Springer, 2025.

2025 Souparni Mazumder, Sanket Biswas, **Alloy Das**, and Josep Lladós. Doc2Graph-X: A Multilingual Graph-Based Framework for Form Understanding. In *Lecture Notes in Computer Science*, pages 257–266. Springer, 2025.

2025 Souparni Mazumder, Sanket Biswas, Aniket Pal, **Alloy Das**, Umapada Pal, and Josep Lladós. Doc2GraphFormer: Bridging Structured Graph Learning with Transformer Attention for Efficient Document Understanding. In *Lecture Notes in Computer Science*, pages 506–522. Springer, 2025.

2025 Koyel Ghosh, Mithun Das, Sumukh Patel, Nilotpal Bhandary, **Alloy Das**, Animesh Mukherjee, Sandip Modha, Debasis Ganguly, Utpal Garain, Sylvia Jaki, and Thomas Mandl. Overview of the HASOC Track at FIRE 2025: Abusive Meme Identification – Shadows Behind the Laughter. In *Forum for Information Retrieval Evaluation (FIRE 2025)*, pages 28–31. ACM, 2025.

2024 **Alloy Das**, Sanket Biswas, Umapada Pal, Josep Lladós, and Saumik Bhattacharya. FastTextSpotter: A High-Efficiency Transformer for Multilingual Scene Text Spotting. In *International Conference on Pattern Recognition (ICPR), Lecture Notes in Computer Science*, pages 135–150. Springer, 2024.

2024 **Alloy Das**, Sanket Biswas, Umapada Pal, and Josep Lladós. Diving into the depths of spotting text in multi-domain noisy scenes. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 410–417. IEEE, 2024.

2024 **Alloy Das**, Sanket Biswas, Ayan Banerjee, Josep Lladós, Umapada Pal, and Saumik Bhattacharya. Harnessing the Power of Multi-Lingual Datasets for Pre-training: Towards Enhancing Text Spotting Performance. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 707–717, 2024.

2022 **Alloy Das**, Rohit Agarwal, Rituparna Singh, Arindam Chowdhury, and Debashis Nandi. Automatic detection of COVID-19 from chest x-ray images using deep learning model. In *AIP Conference Proceedings*, volume 2424, page 040003. AIP Publishing LLC, 2022.

2022 Arindam Chowdhury, Rohit Agarwal, **Alloy Das**, and Debashis Nandi. U-Net Based Optic Cup and Disk Segmentation from Retinal Fundus Images via Entropy Sampling. In *Advanced Computational Paradigms and Hybrid Intelligent Computing: Proceedings of ICACCP 2021*, pages 479–489. Springer, 2022.

Preprint

2026 Qisai Liu, **Alloy Das**, Zhanhong Jiang, Joshua R. Waite, Aditya Balu, Adarsh Krishnamurthy, and Soumik Sarkar. Lighting-aware Unified Model for Instance Segmentation. *arXiv preprint arXiv:2605.20436*, 2026.

2025 Aniket Pal, Sanket Biswas, **Alloy Das**, Ayush Lodh, Priyanka Banerjee, Soumitri Chattopadhyay, Dimosthenis Karatzas, Josep Lladós, and C. V. Jawahar. NoTeS-Bank: Benchmarking Neural Transcription and Search for Scientific Notes Understanding. *arXiv preprint arXiv:2504.09249*, 2025.

Journal Articles

2025 **Alloy Das**, Priyanka Banerjee, Sanket Biswas, Manokaran Kamalakannan, Joydev Chattopadhyay, Dhriti Banerjee, and Tanoy Mukherjee. Tricho-Vision: The use of computer vision in trichotaxonomy for enhancing wildlife conservation of priority species. *Ecological Informatics*, volume 90, page 103161. Elsevier, 2025.

2024 **Alloy Das**, Shivakumara Palaiahnakote, Ayan Banerjee, Apostolos Antonacopoulos, and Umapada Pal. Soft set-based MSER end-to-end system for occluded scene text detection, recognition and prediction. *Knowledge-Based Systems*, volume 305, page 112593. Elsevier, 2024.

2024 Praveen Kumar Pradhan, **Alloy Das**, Amish Kumar, Udayan Baruah, Biswaraj Sen, and Palash Ghosal. SwinSight: a hierarchical vision transformer using shifted windows to leverage aerial image classification. *Multimedia Tools and Applications*, volume 83, pages 86457–86478. Springer, 2024.

2022 Palaiahnakote Shivakumara, **Alloy Das**, KS Raghunandan, Umapada Pal, and Michael Blumenstein. New Deep Spatio-Structural Features of Handwritten Text Lines for Document Age Classification. *International Journal of Pattern Recognition and Artificial Intelligence*, volume 36, page 2252013. World Scientific, 2022.

Research Experience

Iowa State University, Ames, IA, USA.

2025 – Present ***EmbodiedMAE: Multi-modal Masked Autoencoder for 3D Plant Reconstruction***, *Advisor: Prof. Soumik Sarkar.*

A multi-modal masked autoencoder that learns joint representations from RGB images, depth maps, and point clouds for 3D reconstruction and phenotyping of Sorghum plants.

2025 – Present ***Lighting-aware Unified Model for Instance Segmentation***, *Advisor: Prof. Soumik Sarkar.*

Extends the Segment Anything Model (SAM) with a Lighting Convolutional Attention (LCA) module for robust instance segmentation under challenging illumination conditions.

2025 – Present ***Seeing Seizures: Video and EEG Deep Learning for Rodent Seizure Detection and Severity Grading***, *Advisor: Prof. Soumik Sarkar.*

Benchmarked 10 video and 7 EEG models on 525 hours of synchronized video–EEG from 20 rats for seizure detection and Racine severity grading; EEG leads detection, video leads severity grading. Under review. Project page.

Indian Statistical Institute, Kolkata, India.

April. 2023 – Nov. 2024 ***FAST: Font Agnostic Scene Text Editing***, *Advisor: Prof. Umapada Pal .*

We developed a font-agnostic scene text editing framework called FAST. FAST addresses the challenges posed by complex image backgrounds, various font styles, and varying word lengths in existing methods. It generates text in arbitrary styles and locations while preserving a natural appearance.

Jan. 2023 – Feb. 2023 ***Diving into the Depths of Spotting Text in Multi-Domain Noisy Scenes***, *Advisor: Prof. Umapada Pal .*

We examined how noises impact transformer-based scene text spotting systems, specifically utilizing underwater scenes for observation. We introduced the concept of domain-agnostic scene text spotting and provided a validation benchmark tailored to noisy underwater scenes. Additionally, we proposed a novel evaluation benchmarking protocol specific to our underwater text spotting dataset that we annotated for scene text spotting task.

Oct. 2022 – Jan. 2023 ***Harnessing the Power of Intermediate Representations for Domain Adaptation: Towards Enhancing Text Spotting Performance***, *Advisor: Prof. Umapada Pal .*

Our work focuses on domain-adapted scene text spotting, training models to adapt to diverse real-world conditions. We introduce Swin-TESTR, a transformer baseline for accurate and efficient spotting of regular and arbitrary-shaped scene text across multiple domains. Our findings highlight the potential of intermediate representations for improved accuracy and model efficiency in text spotting benchmarks.

Jan. 2022 – Oct. 2023 ***Soft Set-based MSER End-to-End System for Occluded Scene Text Detection, Recognition and Prediction***, *Advisor: Prof. Umapada Pal .*

We introduces a novel Soft Set-based system for text detection, recognition, and prediction in occluded natural scene images. It combines Soft Sets with Maximally Stable Extremal Regions for robust text detection, utilizes a Graph Recurrent Neural Network for grouping text components, and employs a Convolutional Recurrent Neural Network for accurate text recognition and predicting missing characters. Experimental results validate its superior performance compared to existing methods in text-related tasks.

National Institute of Technology, Durgapur, India.

Sept. 2021 – March. 2022 ***Medical Image Segmentation, Glaucoma Detection, and Covid Detection.***, *Advisor: Prof. Debasis Nandi.*

We proposed a novel method to segment medical images and identify diseases. We utilized this method for Glaucoma detection, focusing on identifying condition-related features in retinal fundus images. We then extended the approach to detect COVID-19, with the aim of identifying virus-related abnormalities in chest CT scan imaging data.

Professional Service

Reviewer *The Visual Computer* (Springer), *Scientific Reports* (Nature Portfolio)

Computer skills

Programming	Python, MATLAB, C
Libraries	PyTorch, Keras, Tensorflow, Hugging Face
ML Methods	SAM, Masked Autoencoders, PyTorch DDP, Weights & Biases
HPC & Data	Nova (Iowa State), TACC Vista, NCSA Delta; Unity synthetic data
Documentation	MS Office, LaTeX
Languages	English, Bengali (Native) & Hindi